

Table 3 Error at the apex (mm) and deviation of implant axis (degrees) for sleeve lengths 6 and 7 mm

Sleeve length (mm)	Clearance (μ m)	Deviation ($^{\circ}$)	Total length (mm)													
			16	17	18	19	20	21	22	23	24	25	26	27	28	29
6	50	0.5	0.1	0.1	0.1	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
	80	0.8	0.2	0.2	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
	110	1.1	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.4	0.4	0.5	0.5
	140	1.3	0.3	0.3	0.4	0.4	0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.6	0.6	0.6
	170	1.6	0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.6	0.6	0.6	0.7	0.7	0.7	0.7
	200	1.9	0.4	0.5	0.5	0.5	0.6	0.6	0.6	0.7	0.7	0.7	0.8	0.8	0.8	0.9
	230	2.2	0.5	0.5	0.6	0.6	0.7	0.7	0.7	0.8	0.8	0.8	0.9	0.9	1.0	1.0
	260	2.5	0.6	0.6	0.7	0.7	0.7	0.8	0.8	0.9	0.9	1.0	1.0	1.0	1.1	1.1
	290	2.8	0.6	0.7	0.7	0.8	0.8	0.9	0.9	1.0	1.0	1.1	1.1	1.2	1.2	1.3
	320	3.1	0.7	0.7	0.8	0.9	0.9	1.0	1.0	1.1	1.1	1.2	1.2	1.3	1.3	1.4
	350	3.3	0.8	0.8	0.9	0.9	1.0	1.1	1.1	1.2	1.2	1.3	1.3	1.4	1.5	1.5
	380	3.6	0.8	0.9	1.0	1.0	1.1	1.1	1.2	1.3	1.3	1.4	1.5	1.5	1.6	1.6
	410	3.9	0.9	1.0	1.0	1.1	1.2	1.2	1.3	1.4	1.4	1.5	1.6	1.6	1.7	1.8
7	50	0.4		0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.2	0.2	0.2	0.2
	80	0.7		0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3
	110	0.9		0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.4
	140	1.1		0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.4	0.4	0.5	0.5	0.5	0.5
	170	1.4		0.3	0.4	0.4	0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.6	0.6	0.6
	200	1.6		0.4	0.4	0.4	0.5	0.5	0.5	0.6	0.6	0.6	0.6	0.7	0.7	0.7
	230	1.9		0.4	0.5	0.5	0.5	0.6	0.6	0.6	0.7	0.7	0.7	0.8	0.8	0.8
	260	2.1		0.5	0.5	0.6	0.6	0.7	0.7	0.7	0.8	0.8	0.8	0.9	0.9	0.9
	290	2.4		0.6	0.6	0.6	0.7	0.7	0.8	0.8	0.8	0.9	0.9	1.0	1.0	1.1
	320	2.6		0.6	0.7	0.7	0.8	0.8	0.8	0.9	0.9	1.0	1.0	1.1	1.1	1.2
	350	2.9		0.7	0.7	0.8	0.8	0.9	0.9	1.0	1.0	1.1	1.1	1.2	1.2	1.3
	380	3.1		0.7	0.8	0.8	0.9	1.0	1.0	1.1	1.1	1.2	1.2	1.3	1.3	1.4
	410	3.4		0.8	0.8	0.9	1.0	1.0	1.1	1.1	1.2	1.3	1.3	1.4	1.4	1.5

we focus on the errors made during the manufacturing of such a guide and especially those depending on the way the metal components of the guide interrelate with the osteotomy bur and with the planned position of the implant. However, it should be stressed that the results of this study remain relevant for every implant surgical guide irrespective of the way it is manufactured, 3d-printed or other.

Computer-aided implant surgery is not flawless, even when the guides are manufactured by experts in 3d printing. Three recent systematic reviews have demonstrated errors between the planned and the final implant position. In the study of Schneider et al. [5], the review revealed a mean deviation of 1.07 mm (95% CI 0.76–1.22 mm) at the entry point, 1.6 mm (95% CI 1.26–2 mm) at the apex and a mean angular deviation of the implant axis of 5.3° (95% CI 3.94–6.581). In the study of Van Assche et al. [21], the mean error at the entry point was 0.99 mm (range 0–6.5 mm), at the apex 1.2 mm (range 0–6.9 mm) and the mean axis deviation was 3.81° (range 0–24.9 degrees). In the study of

Tahmaseb et al. [17], the mean error at the entry point was 1.12 mm (maximum 4.5 mm) and 1.39 mm at the apex (maximum 7.1 mm). These reported errors are the summation of flaws in every stage of the procedure in the guided implant surgery: scanning, processing, manufacturing, surgery and not exclusively because of the properties of the mechanical parts of the guide.

A relatively small number of studies are concerned with the errors in the implant positioning due to the mechanical components of the guides. All of them refer to commercially available systems and try to evaluate or even improve these existing systems, elaborating in the mechanical errors of the particular system used [11–16].

Our study is free from the constraints of commercially available systems, but our equations and tables can be used to estimate errors generated by these systems when certain parameters are known. Our study calculates the maximum errors expected by the mechanical components of the computerised implant surgery process, with